SYDE 671 · ASSIGNMENT 3

A3

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PART ONE

1.1 · Source photographs

Two overlapping campus photos are used for the manual mosaic. The three parking-lot photos are used for the automatic mosaic.

Campus: left

Campus: right

Parking lot: left

Parking lot: middle

Parking lot: right

1.2 · Recovering the homography

H maps a point (x, y) in the left photo to (u, v) in the right photo. Setting its last entry to 1 leaves eight unknowns:

$$u = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + 1}$$
$$v = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + 1}$$

Multiplying by the denominator and collecting the unknowns gives two linear equations per point pair:

$$xh_{11} + yh_{12} + h_{13} - uxh_{31} - uyh_{32} = u$$
$$xh_{21} + yh_{22} + h_{23} - vxh_{31} - vyh_{32} = v$$

For eight unknowns, four pairs of (x, y) points are enough to solve for H. Eight pairs are used for better accuracy, giving 16 equations to solve with least squares.

CODE SNIPPET

```
def computeH(im1_pts, im2_pts):
    A = []
    b = []

    # Build equations
    for (x, y), (u, v) in zip(im1_pts, im2_pts):
        A.append([x, y, 1, 0, 0, 0, -u*x, -u*y])
        A.append([0, 0, 0, x, y, 1, -v*x, -v*y])
        b.extend([u, v])

    A = np.array(A, dtype=float)
    b = np.array(b, dtype=float)
    h = np.linalg.lstsq(A, b, rcond=None)[0]
    H = np.append(h, 1).reshape(3, 3)
    return H
```

Selected point coordinates

PAIR	LEFT (X, Y)	RIGHT (U, V)
1	(433.6, 374.4)	(40.7, 296.7)
2	(484.8, 369.4)	(97.1, 302.2)
3	(533.4, 376.6)	(143.2, 317)
4	(560.9, 518.2)	(152.7, 454.4)
5	(520, 531.1)	(113.2, 467.2)
6	(567.5, 502.2)	(158.5, 440.7)
7	(452.2, 448.6)	(48.1, 377.6)
8	(462.5, 577.2)	(42, 510.2)

The resulting H is:

```
[[   1.491956   -0.207044 -515.839816]
 [   0.519659    1.252276 -301.764171]
 [   0.000944   -0.000247    1.      ]]
```

1.3 · Image warping and rectification

The four image corners are mapped through H to find the output bounds. For each output pixel, H inverse gives the source position. Nearest neighbour rounds that position to a pixel:

CODE SNIPPET

```
def warpImageNearestNeighbor(im, H):
    height, width = im.shape[:2]
    x_min, y_min, x_max, y_max = find_bounds(im, H)
    output_height = y_max - y_min + 1
    output_width = x_max - x_min + 1
    warped = np.zeros((output_height, output_width) + im.shape[2:])
    H_inv = np.linalg.inv(H)

    # Inverse warp
    for y in range(warped.shape[0]):
        for x in range(warped.shape[1]):
            source = H_inv @ np.array([x + x_min, y + y_min, 1])
            sx, sy = source[:2] / source[2]
            sx, sy = int(np.floor(sx + 0.5)), int(np.floor(sy + 0.5))
            if 0 <= sx < width and 0 <= sy < height:
                warped[y, x] = im[sy, sx]
    return warped
```

Bilinear interpolation averages the four surrounding pixels: first across the top and bottom rows, then between those two colors:

CODE SNIPPET

```
# Four neighbors
x0, y0 = int(np.floor(sx)), int(np.floor(sy))
x1, y1 = min(x0 + 1, width - 1), min(y0 + 1, height - 1)
dx, dy = sx - x0, sy - y0
# Weighted average
top = (1 - dx) * im[y0, x0] + dx * im[y0, x1]
bottom = (1 - dx) * im[y1, x0] + dx * im[y1, x1]
warped[y, x] = (1 - dy) * top + dy * bottom
```

Full bilinear function

CODE SNIPPET

```
def warpImageBilinear(im, H):
    x_min, y_min, x_max, y_max = find_bounds(im, H)
    output_height = y_max - y_min + 1
    output_width = x_max - x_min + 1
    warped = np.zeros((output_height, output_width) + im.shape[2:])
    H_inv = np.linalg.inv(H)
    height, width = im.shape[:2]

    for y in range(warped.shape[0]):
        for x in range(warped.shape[1]):
            source = H_inv @ np.array([x + x_min, y + y_min, 1])
            sx, sy = source[:2] / source[2]
            if 0 <= sx <= width - 1 and 0 <= sy <= height - 1:
                # Four neighbors
                x0, y0 = int(np.floor(sx)), int(np.floor(sy))
                x1, y1 = min(x0 + 1, width - 1), min(y0 + 1, height - 1)
                dx, dy = sx - x0, sy - y0
                # Weighted average
                top = (1 - dx) * im[y0, x0] + dx * im[y0, x1]
                bottom = (1 - dx) * im[y1, x0] + dx * im[y1, x1]
                warped[y, x] = (1 - dy) * top + dy * bottom
    return warped
```

The H from 1.2 is used for both warpers:

CODE SNIPPET

```
imwarped_nn = warpImageNearestNeighbor(im, H)
imwarped_bil = warpImageBilinear(im, H)
```

Original

Nearest neighbor

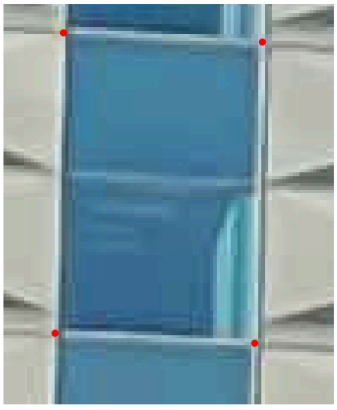
Bilinear

For rectification, the four window corners are matched to four corners of a rectangle using `computeH`. Warping with this H straightens the window.

CODE SNIPPET

```
# Top-left, top-right, bottom-left, bottom-right
corners = np.array([[16.7, 7.8], [74.5, 10.5],
                    [14.3, 94.8], [72.4, 97.6]])
rectangle = np.array([[0, 0], [139, 0],
                      [0, 209], [139, 209]])
H = computeH(corners, rectangle)
imwarped_nn = warpImageNearestNeighbor(im, H)
imwarped_bil = warpImageBilinear(im, H)
```

Original



Nearest neighbor

Bilinear

Runtime in seconds (one run)

IMAGE	NEAREST NEIGHBOUR	BILINEAR
campus warp	6.910 s	13.948 s
campus right window	0.362 s	0.811 s
campus left window	0.723 s	1.705 s

Nearest neighbour looks coarser because it rounds the position, causing jagged edges. Bilinear smooths the neighboring colors, but takes longer because it uses four pixels for each output color.

1.4 · Blending the images

The left photo is warped with `warpImageBilinear` and placed on the same canvas as the right photo. A warped ones image counts where its pixels are present. Dividing the added pixel values by the count averages the overlap.

CODE SNIPPET

```
def createMosaic(im1, im2, H):
    warped1 = warpImageBilinear(im1, H)
    # Coverage: 1 inside the image, 0 outside
    mask1 = warpImageBilinear(np.ones_like(im1), H)
    height1, width1 = warped1.shape[:2]
    height2, width2 = im2.shape[:2]

    # Shared canvas
    x_min1, y_min1, x_max1, y_max1 = find_bounds(im1, H)
    x_min, y_min = min(x_min1, 0), min(y_min1, 0)
    x_max, y_max = max(x_max1, width2 - 1), max(y_max1, height2 - 1)
    mosaic = np.zeros((y_max - y_min + 1, x_max - x_min + 1, 3))
    count = np.zeros_like(mosaic)

    # Add both images to the canvas
    y1, x1 = y_min1 - y_min, x_min1 - x_min
    mosaic[y1:y1 + height1, x1:x1 + width1] += warped1
    count[y1:y1 + height1, x1:x1 + width1] += mask1

    y2, x2 = -y_min, -x_min
    mosaic[y2:y2 + height2, x2:x2 + width2] += im2
    count[y2:y2 + height2, x2:x2 + width2] += 1

    # Average where both images overlap
    return mosaic / np.maximum(count, 1)
```

PART TWO

2.1 · Harris corners and ANMS

The supplied Harris function is used to find the corners. For each corner, ANMS finds stronger corners using `scores[i] < 0.9 * scores[j]`. The Euclidean distance to each of these corners is calculated, and the smallest distance is taken as the radius. The 500 corners with the largest radii are kept.

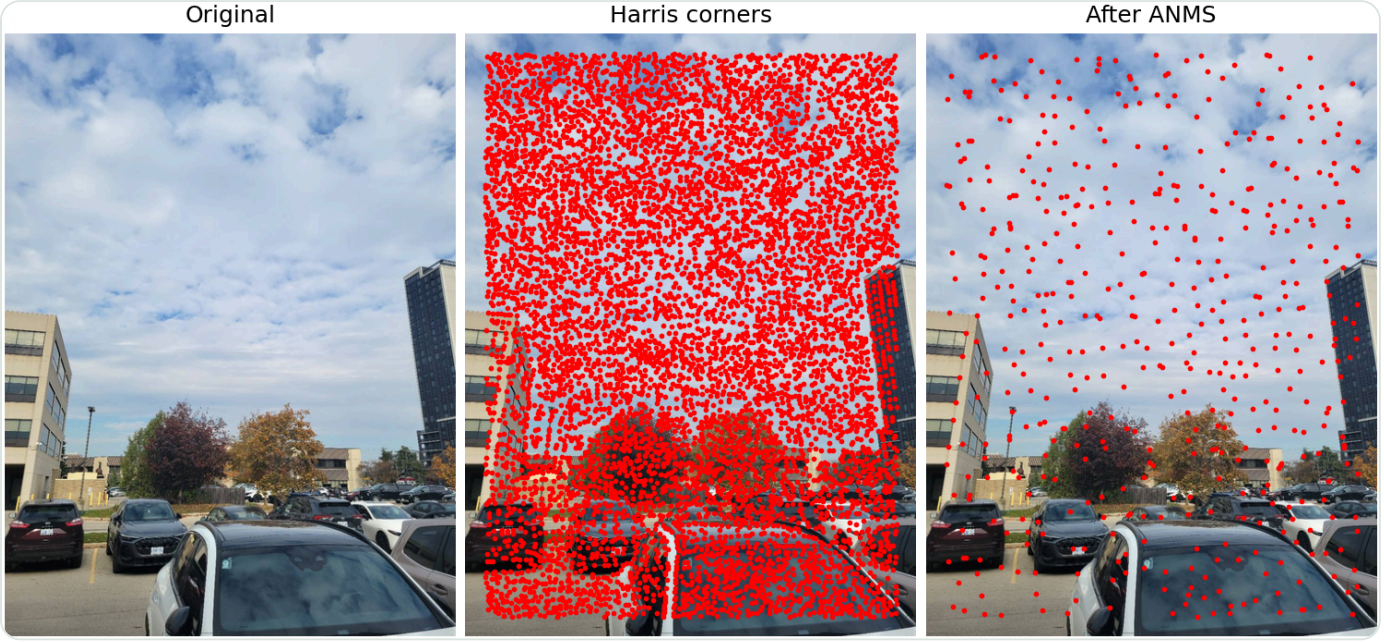
CODE SNIPPET

```
def ANMS(h, coords, num_points=500):
    ys, xs = coords
    points = np.column_stack((xs, ys))
    scores = h[ys, xs]
    radii = np.full(len(points), np.inf)

    # Nearest stronger corner
    for i in range(len(points)):
        x, y = points[i]
        stronger = points[scores[i] < 0.9 * scores]
        if len(stronger) > 0:
            distances = np.sqrt((stronger[:, 0] - x)**2 + (stronger[:, 1] - y)**2)
            radii[i] = distances.min()

    order = np.argsort(radii)[::-1]
    return points[order[:num_points]]
```

IMAGE	HARRIS CORNERS	AFTER ANMS
Left	7,043	500
Middle	7,073	500
Right	6,781	500



2.2 · Feature descriptors

Each corner uses a 40 × 40 patch, blurred with the Gaussian filter from A2. Taking every fifth row and column gives an 8 × 8 descriptor.

We subtract the mean and divide by the standard deviation:

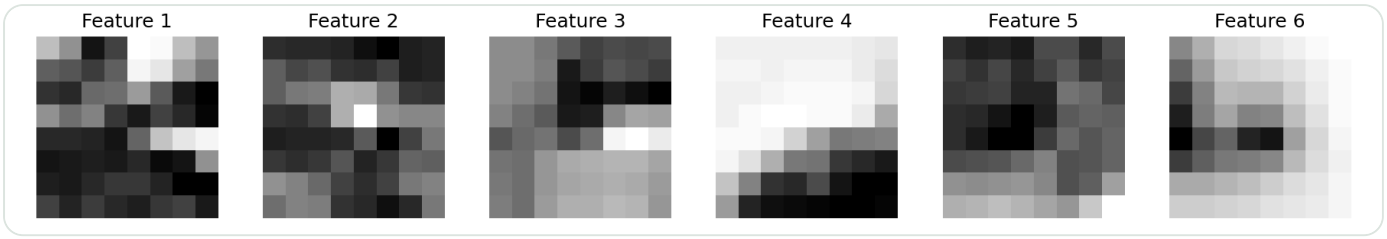
CODE SNIPPET

```
def extractDescriptors(gray, points):
    descriptors = []
    for x, y in points:
        # Blur the 40 x 40 patch
        patch = gray[y - 20:y + 20, x - 20:x + 20]
        patch = gaussian_filter(patch, kernel_size=13, sigma=2, boundary='symm')

        # Sample every fifth pixel
        patch = patch[::5, ::5]

        # Normalize
        patch = patch - patch.mean()
        std = patch.std()
        if std > 0:
            patch = patch / std
        descriptors.append(patch)

    return np.array(descriptors)
```



2.3 · Feature matching

CODE SNIPPET

```
def matchDescriptors(descriptors1, descriptors2, threshold=0.5):
    matches = []
    if len(descriptors2) < 2:
        return np.empty((0, 2), dtype=int)

    for i, descriptor1 in enumerate(descriptors1):
        errors = []
        for descriptor2 in descriptors2:
            error = np.sum((descriptor1 - descriptor2)**2)
            errors.append(error)

        order = np.argsort(errors)
        best_index = order[0]
        second_index = order[1]
        best_error = errors[best_index]
        second_error = errors[second_index]

        if second_error == 0:
            continue

        # Ratio test
        ratio = best_error / second_error
        if ratio < threshold:
            matches.append([i, best_index])

    return np.array(matches, dtype=int).reshape(-1, 2)
```

Descriptors are compared using the sum of squared differences (SSD). A smaller SSD means a closer match. The best match is kept only if its error is less than half the second-best error. If the two errors are similar, the match is rejected.

Left to middle: 128 matches

Right to middle: 55 matches

2.4 · RANSAC and automatic mosaicing

CODE SNIPPET

```
def ransacHomography(points1, points2, iterations=1000, threshold=3):
    best_inliers = []
    for _ in range(iterations):
        sample = np.random.choice(len(points1), 4, replace=False)
        H = computeH(points1[sample], points2[sample])
        if H is None:
            continue

        # Test all matches
        inliers = []
        for i, (point1, point2) in enumerate(zip(points1, points2)):
            prediction = H @ np.append(point1, 1)
            prediction = prediction[:2] / prediction[2]
            error = np.sqrt(np.sum((prediction - point2)**2))
            if error < threshold:
                inliers.append(i)

        if len(inliers) > len(best_inliers):
            best_inliers = inliers

    # Refit using all inliers
    H = computeH(points1[best_inliers], points2[best_inliers])
    return H, np.array(best_inliers)
```

Each trial randomly chooses four matched point pairs and uses `computeH` to calculate H. That H maps every source point to the second image. The Euclidean distance from the mapped point to its matching point is the error; matches below 3 pixels are counted as inliers. This is repeated 1000 times, and the group with the most inliers is kept. H is then calculated again using all the inlier pairs, instead of only the four sampled pairs.

PAIR	MATCHES FROM 2.3	RANSAC INLIERS
left to middle	128	77
right to middle	55	30

Left to middle: 77 inliers

Right to middle: 30 inliers

Recovered homographies

Left to middle:

```
[[   1.356411   -0.152784 -165.4918  ]
 [   0.382292    1.18322  -100.127848]
 [   0.000789   -0.000102    1.      ]]
```

Right to middle:

```
[[  0.51559    0.10958  211.007022]
 [ -0.419744   0.79338   81.873741]
 [ -0.00086   -0.000132   1.      ]]
```

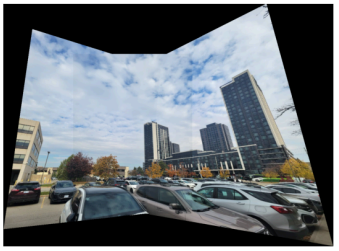
The middle photo stays fixed. The left and right photos are warped using their H matrices, then blended as in 1.4.

Left

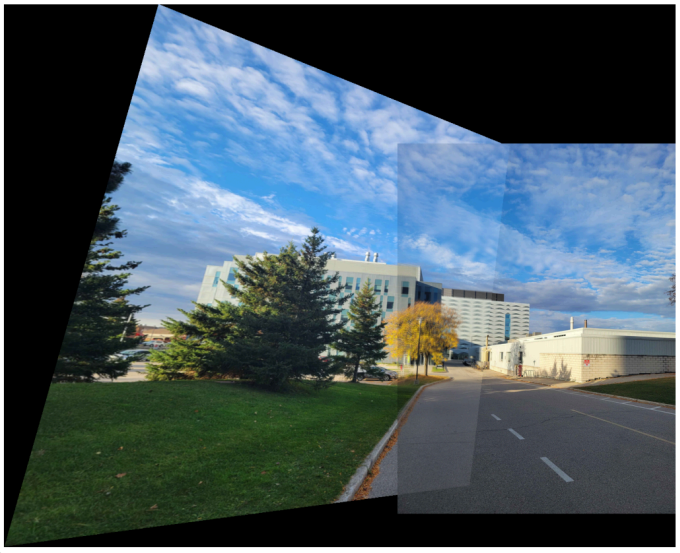
Middle

Right

Part B: automatic



Part A: manual, 2 images



Part B: automatic, 3 images